

PHYSICS 7B

UC BERKELEY

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7B Discussion Solutions, Week 14

Topics

Electromagnetic induction and inductance

Week 14 Discussion Solutions

Electromagnetic induction and inductance

1. A vertical antenna picks up an emf from the horizontal component of $\vec{v} \times \vec{B}$. The maximum emf occurs when the car's velocity is perpendicular to the horizontal component of $\vec{B}$, i.e., when the car moves due east or due west.

The horizontal component of the field is $B_x = B \cos \theta$, where $\theta = 45^\circ$ is the dip angle. The induced emf is

$$\begin{aligned}\mathcal{E} &= B_x \ell v = B \cos \theta \ell v \\ &= (5.0 \times 10^{-5} \text{ T}) \cos(45^\circ) (0.75 \text{ m}) (30 \text{ m/s}) \\ &\approx 8.0 \times 10^{-4} \text{ V} = 0.80 \text{ mV}.\end{aligned}$$

2. (a) As the magnetic flux through the orbit changes, Faraday's law gives a non-zero line integral of the induced electric field around the orbit:

$$\oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}.$$

By symmetry, this induced $\vec{E}$ is tangent to the circular orbit and has the same magnitude everywhere on it. The tangential force $\vec{F} = q\vec{E}$ accelerates the electrons along their orbit. (The radial component of $\vec{B}$ at the orbit simultaneously provides the centripetal force that keeps them on a circle of fixed radius.)

- (b) Take $\vec{B}$ at the orbit to point upward (out of the page when viewed from above) and to be increasing. By Lenz's law the induced current must oppose this increase, i.e., create a magnetic field pointing downward inside the loop. By the right-hand rule, the induced (conventional) current flows clockwise when viewed from above, which means the electrons (negative charges) move *counter-clockwise* when viewed from above.
- (c) For the electrons to gain energy, the induced tangential $\vec{E}$ must point in the direction of their motion. From part (a), this requires $d\Phi_B/dt$ to have a definite sign, hence B must be *increasing* in magnitude (with the appropriate sign so that $\vec{B}$ at the orbit also points in the right direction to bend the electrons inward).
- (d) Write $B(t) = B_0 \sin(\omega t)$ with $\omega = 2\pi(60 \text{ Hz})$. Two conditions must be satisfied simultaneously for the electrons to be accelerated in the same circular sense:
 - B must have the correct sign so that the magnetic force on the electron points toward the center (this fixes one half-cycle, e.g. $B > 0$, i.e. $0 < t < T/2$).
 - $|B|$ must be *increasing* so that the induced $\vec{E}$ accelerates (rather than decelerates) the electrons; for $B > 0$ this means $dB/dt > 0$, i.e. $0 < t < T/4$.

Both conditions are met only during one quarter of each cycle, of duration

$$\frac{T}{4} = \frac{1}{4 \cdot 60 \text{ Hz}} = \frac{1}{240} \text{ s}.$$

- (e) Let r be the (constant) orbit radius and $B_0(t)$ the field at the orbit. The relativistic/non-relativistic relation between momentum and orbit radius is

$$p = qB_0 r \implies \frac{dp}{dt} = qr \frac{dB_0}{dt}.$$

The tangential force on the electron is $F_t = qE$, where the induced electric field at the orbit follows from Faraday's law applied to a circle of radius r :

$$E(2\pi r) = \frac{d\Phi_B}{dt} = \pi r^2 \frac{dB_{\text{avg}}}{dt} \implies E = \frac{r}{2} \frac{dB_{\text{avg}}}{dt},$$

where B_{avg} is the spatial average of B over the area enclosed by the orbit. Newton's second law along the orbit gives

$$\frac{dp}{dt} = qE = \frac{qr}{2} \frac{dB_{\text{avg}}}{dt}.$$

Equating the two expressions for dp/dt :

$$qr \frac{dB_0}{dt} = \frac{qr}{2} \frac{dB_{\text{avg}}}{dt} \implies \frac{dB_0}{dt} = \frac{1}{2} \frac{dB_{\text{avg}}}{dt}.$$

Integrating from $t = 0$ (with $B_0(0) = B_{\text{avg}}(0) = 0$):

$$\boxed{B_0(t) = \frac{1}{2} B_{\text{avg}}(t).}$$

This is why the pole faces are shaped so that the field at the orbit is exactly half of the average field over the enclosed area.

3. (a) Yes. With the rails open, no current flows, so there is no magnetic force on the rod. With no friction either, the rod travels at constant speed.
- (b) With the resistor connected, the changing flux induces an emf

$$\mathcal{E} = \frac{d\Phi_B}{dt} = B \frac{dA}{dt} = B\ell v,$$

so the current is $I = \mathcal{E}/R = B\ell v/R$. The magnetic force on the rod opposes its motion:

$$F = -I\ell B = -\frac{B^2\ell^2 v}{R} = m\dot{v}.$$

Solving with $v(0) = v_0$:

$$v(t) = v_0 e^{-\frac{B^2\ell^2}{mR} t}.$$

The rod decelerates exponentially; its kinetic energy is dissipated as heat in R .

- (c) With a constant current I , the magnetic force is constant, $F = I\ell B$:

$$m\dot{v} = I\ell B \implies v(t) = \frac{I\ell B}{m} t.$$

- (d) With a constant emf $\mathcal{E}_0$, the rod's motion induces a back-emf $B\ell v$, so the net emf driving the current is $\mathcal{E}_0 - B\ell v$:

$$m\dot{v} = \frac{\mathcal{E}_0 - B\ell v}{R} \ell B \implies v(t) = \frac{\mathcal{E}_0}{B\ell} \left(1 - e^{-\frac{B^2\ell^2}{mR} t}\right).$$

- (e) In case (c), no terminal speed: v grows linearly without bound. In case (d), yes: the terminal speed is $v_\infty = \mathcal{E}_0/(B\ell)$.
4. (a) Choose the area normal into the page (same direction as $\vec{B}$). Then $\Phi_B > 0$ and increasing. By Lenz's law, the induced current opposes this increase by producing a magnetic field out of the page inside the loop. By the right-hand rule, the induced current therefore flows *counter-clockwise*.
 - (b) Faraday's law (the area is constant):

$$|\mathcal{E}| = \frac{d\Phi_B}{dt} = A \frac{dB}{dt} = \pi r^2 a.$$

The emf is drawn as a battery whose positive terminal is on the counter-clockwise side of the negative terminal (so that it would, on its own, drive a CCW current).

(c) Kirchhoff's loop rule: $\mathcal{E} = V_C + V_R = Q/C + R\dot{Q}$, so

$$\dot{Q} + \frac{Q}{RC} = \frac{\mathcal{E}}{R}.$$

With $Q(0) = 0$:

$$Q(t) = C\mathcal{E} \left(1 - e^{-t/RC}\right) \implies V_C(t) = \frac{Q(t)}{C} = \pi r^2 a \left(1 - e^{-t/RC}\right).$$

(d) At $t = 0^+$ the capacitor is uncharged, so it acts like a wire. The equivalent circuit is just the emf source $\mathcal{E}$ in series with R , and the power dissipated by the resistor is

$$P(0) = \frac{\mathcal{E}^2}{R} = \frac{(\pi r^2 a)^2}{R}.$$

5. The flux through the N -turn loop is

$$\Phi_B(t) = N \int \mathbf{B} \cdot \hat{n} dA = N\pi r^2 B_0 e^{-t/\tau}.$$

The induced emf:

$$\mathcal{E}(t) = -\frac{d\Phi_B}{dt} = \frac{N\pi r^2 B_0}{\tau} e^{-t/\tau}.$$

Instantaneous power dissipated in the resistance R :

$$P(t) = \frac{\mathcal{E}^2(t)}{R} = \frac{N^2 \pi^2 r^4 B_0^2}{\tau^2 R} e^{-2t/\tau}.$$

Total energy dissipated up to time t :

$$E(t) = \int_0^t P(t') dt' = \frac{N^2 \pi^2 r^4 B_0^2}{2\tau R} \left(1 - e^{-2t/\tau}\right).$$

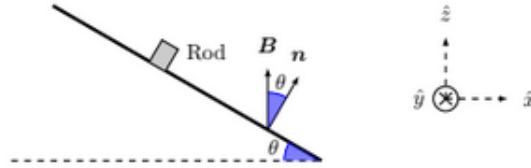
$$\text{As } t \rightarrow \infty, E_\infty = \frac{N^2 \pi^2 r^4 B_0^2}{2\tau R}.$$

6. Solution:

Solution 21.2:

1. There are many steps here. First we will use Faraday's Law to find the EMF generated in the wire. Second, we will use Ohm's law to determine the current in the wire. Third, we will use the Lorentz force law to determine the magnetic force on the rod. Next, we will draw a free-body diagram and decompose forces to find the equation of motion for the rod. Finally, we will apply the condition for terminal velocity (no acceleration) to find the terminal velocity of the rod.

Let's consider a side view of the situation.



Suppose that the distance between the bottom of the rails and the rod is $x(t)$. Then the magnetic flux is

$$\Phi_B(t) = \int \mathbf{B} \cdot \hat{n} dA = BA(t) \cos \theta = BLx(t) \cos \theta.$$

By Faraday's law,

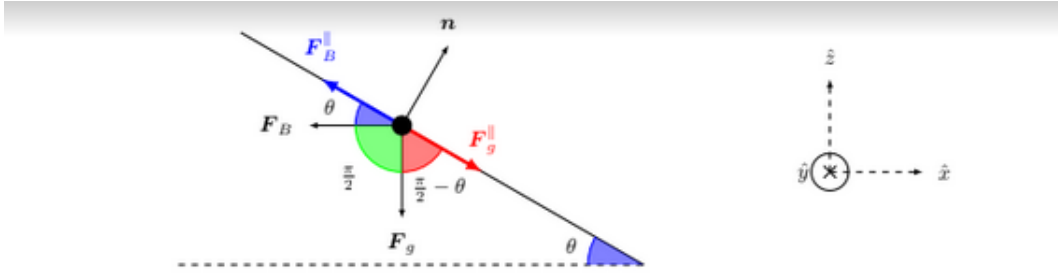
$$\mathcal{E} = -\frac{d\Phi_B(t)}{dt} = -BL \cos \theta \frac{dx}{dt} = -BL \cos \theta v.$$

The flux through the loop is decreasing so, by Lenz's law, the current created by the EMF wants to oppose the change in flux and thus will be directed so that the induced magnetic field points along $\hat{n}$. This means the current must flow towards us through the rod, i.e. $\mathbf{I} = \frac{\mathcal{E}}{R} \hat{y}$ (note that $\mathcal{E}$ is itself a negative quantity, so this points along the negative $\hat{y}$ direction).

Knowing the current, we can calculate the force on the rod:

$$\mathbf{F}_B = \mathbf{I} \ell \times \mathbf{B} = \ell I \hat{y} \times B \hat{z} = -\ell I B \hat{x} = \ell \frac{-BL \cos \theta v}{R} B \hat{x} = -\frac{B^2 \ell^2 \cos \theta}{R} v \hat{x}.$$

We now want to use a free-body diagram to find out where the rod will go.



There are three relevant forces: the magnetic force $\mathbf{F}_B$, gravity $\mathbf{F}_g = -mg\hat{z}$ and the normal force $\mathbf{N}$. The normal force will exactly cancel out the components of the magnetic and gravitational forces that try to pull the rod into the rails. To find the motion of the rod, we only need to know the components of the forces parallel to the rails. Since the normal force is perpendicular to the rails, we don't have to worry about it at all. Therefore the forces we care about are just those parallel to the rails:

$$F_\parallel = m \frac{dv}{dt} = F_B^\parallel + F_g^\parallel.$$

To find the parallel components, we simply project the force vectors along the rails. From the geometry of the diagram,

$$F_\parallel = m \frac{dv}{dt} = F_B^\parallel + F_g^\parallel = F_B \cos \theta + F_g \cos \left(\frac{\pi}{2} - \theta \right) = -\frac{B^2 \ell^2 \cos \theta}{R} v \cos \theta + mg \sin \theta.$$

The rod reaches terminal velocity when $\frac{dv}{dt} = 0$, so at

$$0 = -\frac{B^2 L^2 \cos \theta}{R} v \cos \theta + mg \sin \theta.$$

Solving for v yields

$$v = \frac{mgR \sin \theta}{B^2 L^2 \cos^2 \theta}.$$

7. Solution:

a) Applying Kirchhoff's loop rule we get:

$$V_0 - L \frac{dI}{dt} - RI = 0 \Leftrightarrow \frac{dI}{dt} = -\frac{R}{L}I + \frac{V_0}{L} \quad (1)$$

Where we have manipulated the equation to bring it to the form it appears in the equation sheet.

The solution given is:

$$I(t) = \frac{V_0}{R}(1 - e^{-At}) + I(0)e^{-At} = \frac{V_0}{R}(1 - e^{-\frac{t}{\tau}}), \quad (2)$$

where we have used that the initial current $I(0) = 0$ and that $A = \frac{R}{L} = \tau^{-1}$ is the inverse of the time constant.

b) This part asks us to justify our assumption that $I(0) = 0$. When the switch is open there can be no current, and because there is an inductor in the system, the current cannot change discontinuously. Therefore, the current is still zero immediately after closing the switch.

c) As found in part a), $\tau = \frac{L}{R} = 0.01s$.

d) The current is monotonically increasing in time, starting at zero and asymptoting its maximum value at infinity. Plugging in $t = \infty$ in equation (2), we see that $I_{max} = \frac{V_0}{R} = 1A$.

This same answer can also be reached by arguing that at late times, when the system reaches a steady state and the current becomes constant, the inductor acts as a wire, so by Ohm's Law $V_0 = I_{max}R$, which implies $I_{max} = \frac{V_0}{R}$.

e) The power delivered by the battery is always given by $P = IV_0$. Therefore, at the maximum current we have $P_{max} = I_{max}V_0 = \frac{V_0^2}{R} = 10W$.

f) The energy stored in an inductor is given by $U_L = \frac{1}{2}LI^2$, which means that the power (rate of energy being stored) is given by $P_L = \frac{dU_L}{dt} = LI \frac{dI}{dt} = 0$, since the current becomes constant at late times.